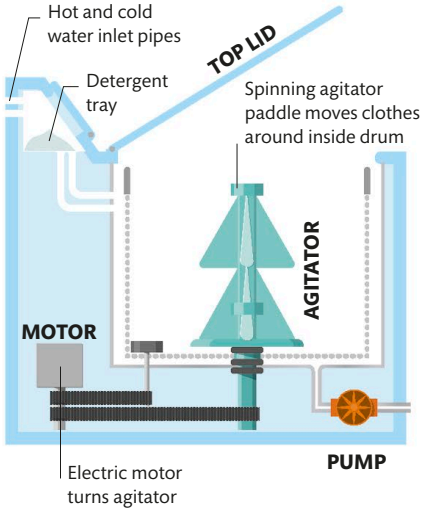
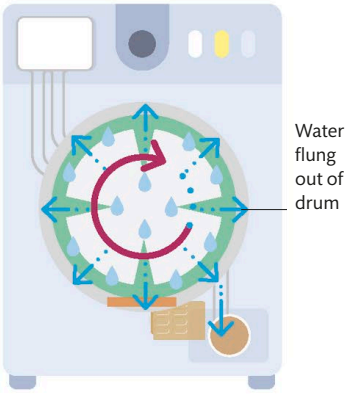


Top loaders

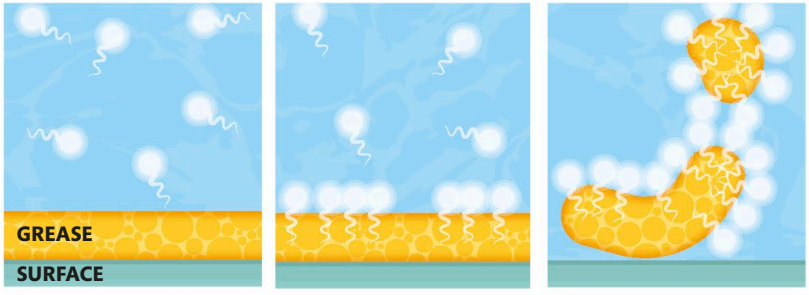
These machines also possess an outer and inner drum, but neither drum moves during the washing cycle. Instead, the clothes and water-detergent mix are stirred thoroughly by a large rotating central agitator, which is powered by an electric motor. The same motor rotates the inner drum during the spin cycle to remove water from the load.



4 Rapid spin and drain
The motor rotates the inner drum at high speeds (300–1,800 rpm), flinging water out of the inner drum. Hot air may be blown into the drum to help the load dry.

Detergents

Most stains and dirt can be removed with hot water alone, but other particularly oily or greasy deposits require chemical help. Detergent molecules contain one acidic end, which is hydrophilic (attracted to water molecules), and a long hydrocarbon chain at the other, which is attracted to oil. Together, they attach to stains and help lift oil and grease away from fabrics.

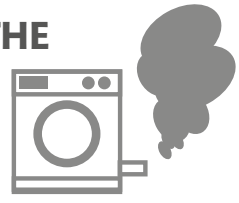


1 Detergent released
Detergent dissolves, and its molecules are mixed in water in the washing-machine drum, coming into contact with oil and grease stains in fabrics.

2 Attaching to dirt
Repelled by water but attracted to oil, one end of the detergent molecules attaches to the stain. Multiple detergent molecules build up, engulfing the stain.

3 Dirt removed
Agitation during the washing cycle and the pull of the hydrophilic end of the detergent molecules lifts the oil or grease out of the fabric, to be rinsed away.

SOME WASHING MACHINES IN THE 1920S WERE POWERED BY A GAS-FUELED ENGINE THAT BELCHED EXHAUST FUMES



Tumble dryers

Wet clothes are placed in a tumble dryer's large drum, which turns slowly, powered by a belt-driven motor. On many models, the drum changes direction frequently to prevent clothes from bunching up. The laundry is tumbled from the drum's top to bottom, through dry, warm air blown into the drum by a fan, and heated by an electric element. The warm, moist air is then carried out through a vent—in some dryers, it first passes over a heat exchanger to extract heat energy.

